

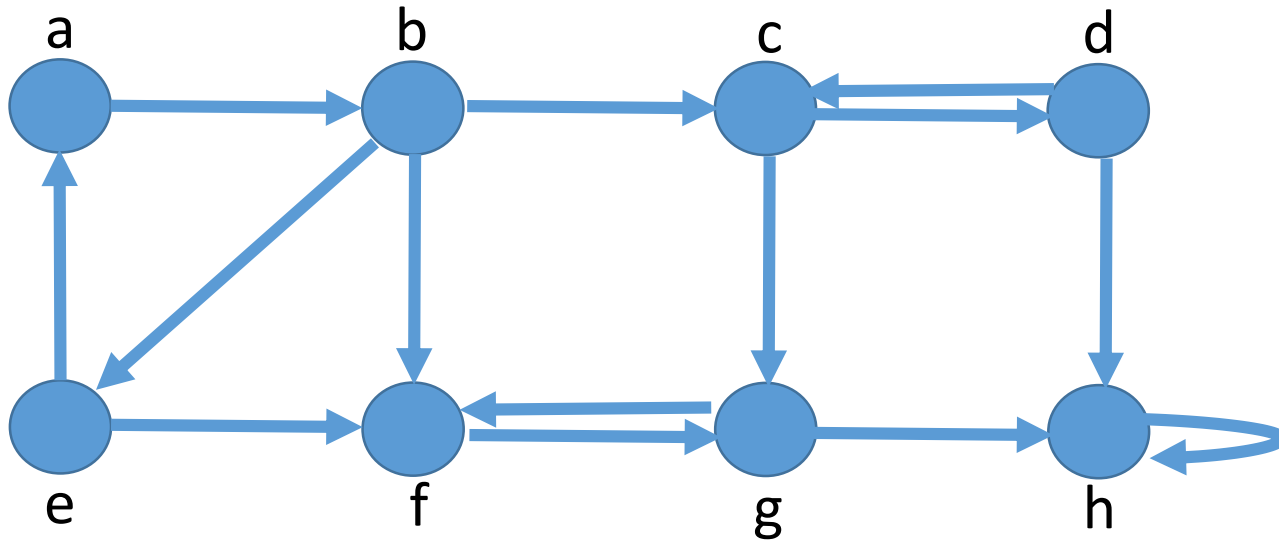
# Finding Strongly Connected Components

## Definition:

A *strongly connected component* of a digraph  $G = (V, E)$  is a maximal set of vertices  $U \subseteq V$  such that for every pair of vertices  $u, v \in U$ , there exists a (directed) path from both  $u$  to  $v$  ( $u \rightsquigarrow v$ ) and from  $v$  to  $u$  ( $v \rightsquigarrow u$ ).

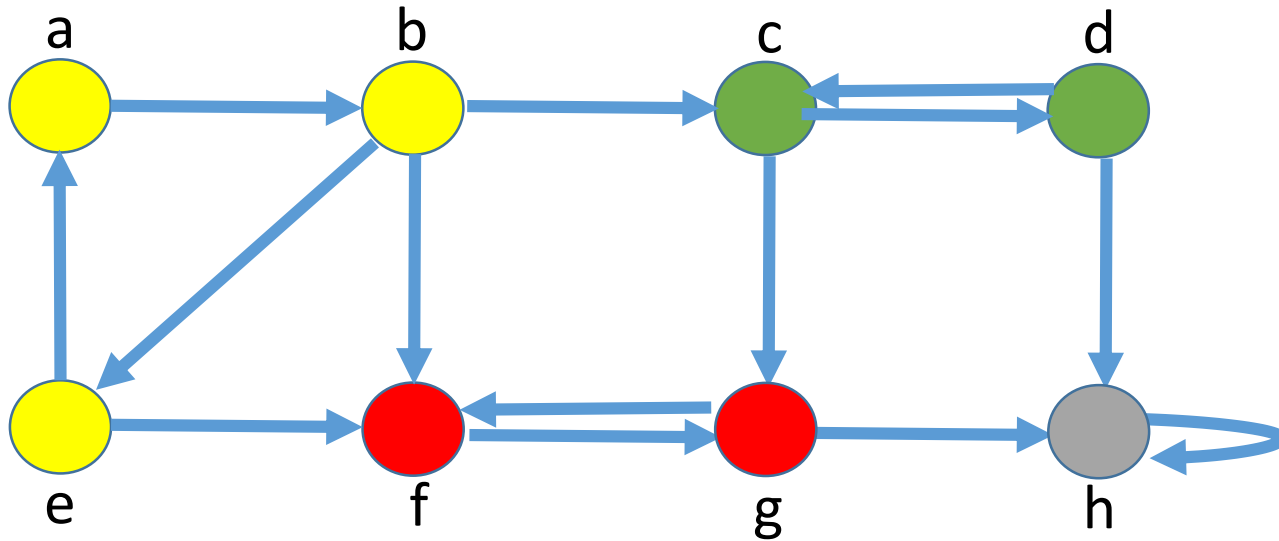
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1. DFS(G)

2.  $G^T = \text{ComputeTranspose}(G)$

(sort V by decreasing f values)

3. DFS( $G^T$ )

4. OutputForest( $G^T$ )

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Claim (SCC algorithm correctness):

Every set of vertices in a tree  $T$  of the DFS-forest of  $G^T$  is a strongly connected component of  $G$ .

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Let  $T$  be a tree of the DFS-forest of  $G^T$ . We show that  $T$  is a strongly connected component of  $G$ .

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## Proof:

Let  $T$  be a tree of the DFS-forest of  $G^T$ . We show that  $T$  is a strongly connected component of  $G$ .

By definition of scc, we need to show that for every pair of vertices  $u, v \in T$ , it holds that:

$u \rightsquigarrow v$  and  $v \rightsquigarrow u$ , and also that  $T$  is maximal.



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## Proof (cont.):

Recall that we need to show that for every pair of vertices  $u, v \in T$ , it holds that  $u \rightsquigarrow v$  and  $v \rightsquigarrow u$  and also that  $T$  is maximal.

Let  $t \in T$  be the root of  $T$ .

1

2

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To show 1, it is enough to show that for every  $u \in T$  it holds that  $u \rightsquigarrow t$  and  $t \rightsquigarrow u$  (since then,  $u \rightsquigarrow t \rightsquigarrow v$  and vice versa).

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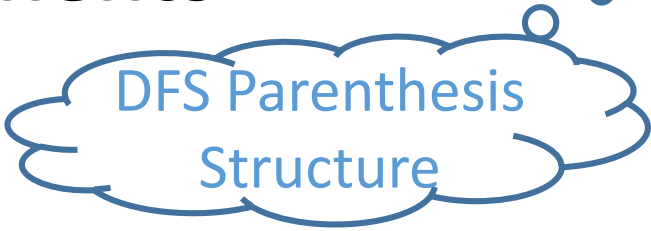
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Since  $T$  is a tree in the DFS-forest of  $G^T$ ,  $t$  its root and  $u$  a node in  $T$ :  $t \rightsquigarrow u$  in  $G^T$ . By  $G^T$  definition, we then have:  $u \rightsquigarrow t$  in  $G$ .

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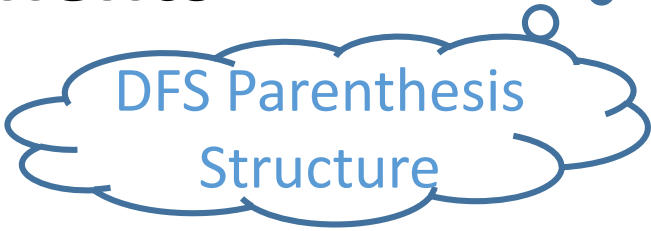
DFS Parenthesis  
Structure

## Proof (cont.):

Consider two vertices  $t$  and  $u$  in  $G$ . There are 4 possibilities in  $\text{DFS}(G)$ :

- (1)  $d[t] \leq d[u] \leq f[u] \leq f[t] \Rightarrow u$  is a descendant of  $t$
- (2)  $d[u] \leq d[t] \leq f[t] \leq f[u] \Rightarrow t$  is a descendant of  $u$
- (3)  $d[t] \leq f[t] \leq d[u] \leq f[u] \Rightarrow$  no path from  $t$  to  $u$
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In our case, we know that  $f[t] > f[u]$  (since  $t$  is the root of  $T$ , it appears before  $u$  in the sorted list of vertices) and also that  $u \rightsquigarrow t$  in  $G$ .

Therefore, only possibility (1) remains. Thus, we have:  $t \rightsquigarrow u$  in  $G$ .

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We now show 2: that  $T$  is maximal.

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We show that for any  $v \notin T$ , either  $v \rightsquigarrow t$  or  $t \rightsquigarrow v$ .

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Let  $v \notin T$ . Since every vertex appears in some tree output, there exists  $T'$  in the DFS forest of  $G^T$  such that  $v \in T'$ .



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So no vertex can be added to  $T$ , therefore,  $T$  is maximal.